

Sets in Python

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Introduction to Python Sets

- ▶ Set is a collection of non repetitive items
- ▶ Sets store multiple items in a single variable
- ▶ Unordered, unindexed, and unchangeable (but items can be added/removed)
- ▶ No duplicate values

Creating a Set

- ▶ Syntax:

```
newset = {"apple", "banana", "cherry"}  
print(newset)
```

- ▶ Using type()

- ▶ Using len()

- ▶ `b = set()` //Creating an empty set

- ▶ Using Set constructor to create a set

```
newset = set(("apple", "banana", "cherry"))
```

Adding Items to a Set

- ▶ Add a single item:

```
newset.add("orange")
```

- ▶ Add multiple items:

```
newset.update([55,66,77])
```

```
newset.add((8,10,9,5)) //to add a list
```

Removing Items from a Set

- ▶ Remove an item using `remove()`:
`newset.remove("banana")`
- ▶ Remove an item using `discard()`:
`newset.discard("banana")`
- ▶ Remove a random item:
`removed_item = thisset.pop()`
- ▶ Empty the set:
`thisset.clear()`

Joining Sets

- ▶ Intersection of two sets:

```
set1 = {"a", "b", "c"}
```

```
set2 = {1, 2, 3}
```

```
set3 = set1.intersection(set2)
```

- ▶ Union of two sets:

```
set3 = set1.union(set2)
```

```
print(set3)
```

```
set5=set1.union(set2,set3,set4)
```

- ▶ Using update():

```
set1.update(set2)
```

- ▶ Using difference()

```
set5=set1.difference(set2)
```

Programs

- ▶ Write a program to input 5 numbers from the user and display only unique numbers

Summary

- ▶ Sets are unordered, unchangeable (except adding/removing items), and do not allow duplicates
- ▶ Useful for operations like union, intersection, and difference